

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1819

Potash Junction 138 kV
November 02, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1819. The Interconnection Customer proposes a 199.0 MW solar photovoltaic facility interconnecting at the Potash Junction 138 kV substation in the Pennsylvania area. The request was received on July 19, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 584 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$658,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Coopers Mills – Eagle 138.0 kV	Lighthouse Hill – Aes Beaver Vly Partners Beaver Valley 345.0 kV L	102.5	194	24.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Coopes Mills – Eagle 138.0 kV L	Reconductor line	\$1,286,000	51% (\$658,000)
	Total allocated to project		\$658,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.